

BU7160 Individual Assignment

This individual assignment is a data analysis project. You are given a data set to demonstrate your analytics skills. This dataset contains sales transaction records for an electronics company over a one-year period, spanning from September 2023 to September 2024. It includes detailed information about customer demographics, product types, and purchase behaviours. This dataset is adapted from the Customer purchase behavior - Electronic Sales Data which is publicly available on Kaggle.com.

Below is the information about the data set for this assignment, and the data set is available in Excel format on Blackboard.

Key Features:

- Customer ID: Unique identifier for each customer.
- Age: Age of the customer (numeric)
- Gender: Gender of the customer (Male or Female)
- Loyalty Member: (Yes/No) (Values change by time, so pay attention to who cancelled and who signed up)
- Product Type: Type of electronic product sold (e.g., Smartphone, Laptop, Tablet)
- SKU: a unique code for each product.
- Rating: Customer rating of the product (1-5 stars) (Should have no Null Ratings)
- Order Status: Status of the order (Completed, Cancelled)
- Payment Method: Method used for payment (e.g., Cash, Credit Card, Paypal)
- Total Price: Total price of the transaction (numeric)
- Unit Price: Price per unit of the product (numeric)
- Quantity: Number of units purchased (numeric)
- Purchase Date: Date of the purchase (format: YYYY-MM-DD)
- Shipping Type: Type of shipping chosen (e.g., Standard, Overnight, Express)
- Add-ons Purchased: List of any additional items purchased (e.g., Accessories, Extended Warranty)
- Add-on Total: Total price of add-ons purchased (numeric)

As a business analyst, you can freely explore the data set and find insights through data analysis. You need to complete a report to summarize all insights supported by data analysis and discussions. In your report, you should study and answer **up to three research questions**, namely you can focus on one research question in detail, or study 2 or 3 different questions in your report. The page limit is 20 pages, including everything, i.e., figures, tables, references and appendix. It should be produced in a serif font, such as Times New Roman, size 12 and single line spacing. The assignment should be completed independently.

A typical report should include the following key parts, and you are free to decide the structure of your own report:

- **Introduction:** Briefly introduce the questions you are studying in the report with justifications, i.e., through a brief literature review or background introduction.
- **Analysis:** Details of the analysis tools you use, new variables you generate, process for analysis and the results
- **Discussion/Conclusion:** discuss the insights from your analysis and the limitations
- **Reference:** list any reference you used in the report (using Harvard style)
- **Appendix (optional):** anything else that you would like to include to support your analysis

Submission is via Blackboard. Please only submit one single pdf file including everything. Do not include name or student number, as the marking is anonymous.